

CLINICAL AND POPULATION STUDIES

Efficacy of the Shear Stress Pattern Computed in Coronary Artery Reconstruction Models With and Without Side Branches in Predicting Plaque Progression

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BACKGROUND: The incorporation of side branches in vessel geometry influences wall shear stress (WSS) distribution. However, complete vessel reconstruction is time-consuming, and there is no evidence that its WSS estimations better predict atherosclerotic disease progression compared with the output of the conventional single-vessel reconstruction (SVR).

METHODS: Patients who had baseline and 1-year follow-up intravascular ultrasound imaging (n=40 vessels), and patients with neoatherosclerotic lesions (n=13 vessels) on optical coherence tomography were included. All the studied vessels had at least one side branch with a diameter >1 mm; 3-dimensional complete vessel reconstruction and SVR were performed, and the time-averaged WSS and multidirectional WSS were computed. The performance of both methods in predicting disease progression in intravascular ultrasound and optical coherence tomography models was assessed.

RESULTS: The incorporation of side branches in 3-dimensional geometry resulted in lower minimum predominant time-averaged WSS in the intravascular ultrasound (1.09 versus 1.58 Pa, $P<0.001$) and optical coherence tomography–based reconstructions (0.68 versus 1.33 Pa, $P<0.001$) and influenced the multidirectional WSS distribution. In native segments, complete vessel reconstruction–derived WSS metrics demonstrated superior predictive performance for disease progression—defined as lumen area reduction and plaque burden increase—compared with SVR, as evidenced by improved out-of-sample accuracy (leave-one-out information criterion: 429 versus 551), discrimination (C statistic: 0.725 versus 0.651), calibration (Brier score: 0.172 versus 0.226), and explained variance (27.8% versus 20.7%). Consistent findings were observed in stented segments, where complete vessel reconstruction–derived WSS metrics more accurately predicted neointimal proliferation than SVR-derived metrics.

CONCLUSIONS: Incorporating side branches into vessel reconstruction influences WSS distribution and enables more accurate prediction of atherosclerotic disease progression in native and stented segments than SVR.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: atherosclerotic disease progression ■ computational fluid dynamics ■ intravascular ultrasound ■ optical coherence tomography ■ shear stress

Experimental and clinical studies suggest that the local hemodynamic forces distribution is an instigator of vascular inflammation and vulnerable plaque

formation.^{1–4} Traditionally, in clinical studies, vessel 3-dimensional (3D) reconstruction is performed by fusing high-resolution intravascular imaging and angiographic

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Nonstandard Abbreviations and Acronyms

3D	3-dimensional
CFI	cross-flow index
CVR	complete vessel reconstruction
IVUS	intravascular ultrasound
NC	necrotic core
NIRS	near-infrared spectroscopy
OCT	optical coherence tomography
OSI	oscillatory shear index
PB	plaque burden
QCA	quantitative coronary angiography
ROPE	region of practical equivalence
RRT	relative residence time
SOI	segment of interest
SVR	single vessel reconstruction
TAWSS	time-averaged wall shear stress
TransWSS	transverse wall shear stress
WSS	wall shear stress

data. However, the methodologies developed for this purpose make assumptions in the coregistration of the 2 modalities and the estimation of the orientation of the intravascular images onto the vessel centerline. In addition, most methodologies do not incorporate side branches in the 3D model despite their impact on flow patterns. Studies comparing wall shear stress (WSS) distribution in models with and without side branches showed that the WSS was lower in the complete vessel reconstructions (CVR), which was attributed to the blood flow diversion into the side branches that resulted in a reduction of the flow in the main vessel.^{5,6} Considering these findings, a recent consensus document on the assessment of WSS in coronary artery reconstructions advocates the incorporation of the side branches in 3D vessel geometry.⁷

However, the methodologies proposed for CVR from intravascular imaging and X-ray data make approximations in the fusion of the 2 data sets, which is time-consuming, and the computational fluid dynamic analysis of the generated complex geometries is computationally expensive. Moreover, it is unclear whether the WSS estimations of these approaches outperform the output of the computational fluid dynamics analyses, performed in models of the main vessel, in predicting plaque progression.⁸

The aims of the present study are to compare the output of the coregistration algorithms included in methodologies proposed for main vessel or CVR from the fusion of intravascular imaging and angiographic data, examine the WSS distribution in models generated from these approaches, and evaluate the performance of the computed WSS in predicting atherosclerotic disease progression.

What Are the Clinical Implications?

Local hemodynamic forces critically influence plaque evolution and vulnerable plaque formation. Conventional methods that fuse intravascular imaging with angiography reconstruct vessel anatomy but often exclude side branches, which can significantly alter flow patterns. Recent techniques addressing this limitation frequently rely on geometric approximations, are time-consuming, and require substantial computational resources. Moreover, it remains unclear whether wall shear stress estimates from models including side branches outperform those from main vessel-only reconstructions in predicting plaque progression. In this study, 40 coronary vessels with serial intravascular ultrasound were reconstructed using 2 approaches: single-vessel reconstruction and complete vessel reconstruction, the latter incorporating side branches ≥ 1 mm. Computational flow simulations were performed to assess the ability of wall shear stress to predict plaque progression. Complete vessel reconstruction resulted in lower wall shear stress values and altered multidirectional wall shear stress patterns compared with single-vessel reconstruction. Importantly, complete vessel reconstruction enabled more accurate prediction of plaque progression and showed stronger associations with changes in lumen area and plaque burden at follow-up. Despite greater geometry complexity and computational cost, incorporating side branches improves physiological realism and enhances the predictive value of 3-dimensional vessel models, supporting their role in risk stratification and detection of vulnerable plaques.

METHODS

Data Availability

The data that support the findings of this study are available from the corresponding author on reasonable request.

Study Design

Comparison of the 2 reconstruction approaches was performed in models reconstructed from the fusion of X-ray angiography with intravascular ultrasound (IVUS) or optical coherence tomography (OCT) data. The IVUS cohort included data from the IBIS-4 (Integrated Biomarkers and Imaging Study-4; URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT00962416) and PACMAN-AMI (Effects of the PCSK9 Antibody Alirocumab on Coronary Atherosclerosis in Patients With Acute Myocardial Infarction; URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT03067844) studies, 2 multimodality intravascular imaging studies that examined the effects of aggressive lipid-lowering therapy on plaque evolution at 13 and 12 months follow-up, respectively, in patients presenting with an acute myocardial infarction. The inclusion and exclusion criteria, and the study design, and end points have been reported elsewhere.^{9,10} In the present analysis, we have only included patients who received statin monotherapy and had at least a coronary artery segment

that could be reconstructed from IVUS and angiographic data at baseline and follow-up, and included a side branch with a diameter >1 mm. Short segments (<15 mm), cases with IVUS hang-up during pullback, and those with extensive foreshortening or overlapping on angiography were excluded. The recruited patients provided written consent; both studies were performed in accordance with the Declaration of Helsinki. We have provided a Strengthening the Reporting of Observational Studies in Epidemiology checklist in the [Supplemental Material](#).

The OCT cohort consisted of the patients recruited in a study that examined the implication of the WSS distribution on neoatherosclerotic lesion formation. The study details and analysis protocol have been described previously.¹¹ In the present study, we analyzed angiographic and OCT imaging data only from patients who had a stented segment with at least 1 side branch with a diameter >1 mm.

Data Acquisition and Image Analysis

In IBIS-4 study, IVUS imaging was performed with a 20MHz IVUS catheter (Eagle Eye, Volcano Cooperation, Rancho Cordova, CA) using a motorized pullback speed of 0.5 mm/s after administration of intracoronary nitroglycerin. IVUS imaging in the patients included from the PACMAN-AMI study was performed with a combined 3.2F near infrared spectroscopy (NIRS)-IVUS catheter (40 MHz, INSIGHT TVC-C195-22, Infraredx, a Nipro Company, Burlington, MA) that was pulled back by a motorized pullback device at the speed of 0.5 mm/s after administration of intracoronary nitroglycerin.

The IVUS images acquired at baseline and follow-up in the 2 studies were reviewed by an expert analyst who identified the most proximal and distal side branches that were visible in both time points on IVUS and in coronary angiography, and these branches were used to define the proximal and distal ends of the segment of interest (SOI), respectively. Only SOIs that included at least 1 side branch >1 mm were included in the present analysis.

In IBIS-4 study, the end-diastolic frames of the SOI were automatically detected by Volcano's proprietary technology, while the NIRS-IVUS data from the PACMAN-AMI study were processed using an in-house deep-learning methodology that enables retrospective end-diastolic frame detection.¹² The lumen and external elastic membrane borders were annotated in the end-diastolic frames and plaque analysis was performed in the IBIS-4 imaging data using Volcano's virtual histology technology—that automatically detects 4 tissue types: fibrous tissue (green), fibro-fatty (light green), dense calcium (white), and necrotic core (NC, red)—and in NIRS-IVUS using a deep-learning technology that has been trained and validated against histology and identifies 3 tissue types: fibrotic, calcific, and NC.^{13,14} In addition, in frames portraying side branches, the direction of the branch was marked with a vector.

The OCT imaging and segmentation details are provided in the [Supplemental Material](#). All analyses were performed using the QCU-CMS (Quantitative Coronary Ultrasound-Cardiovascular Measurement System) software (Version 4.69, Leiden, University Medical Center, the Netherlands).

Coronary Artery Reconstruction

The SOI in native and stented vessels was reconstructed using 2 methodologies: the first—single-vessel reconstruction (SVR) method, where only the main vessel was reconstructed,

whereas the second—CVR which incorporated side branches with a diameter >1 mm in the final geometry (Figure 1).

SVR Method

This approach includes (1) the extraction of the lumen centerline for the SOI from 2 end-diastolic angiographic projections ($>25^\circ$ apart). This process involves 2-dimensional quantitative coronary angiography (QCA) of the SOI in these projections and generation of the 3D-QCA model and its centerline using the QAngioXA 3D Research Edition 1.0 software (Medis Medical Imaging Systems, Leiden, the Netherlands). (2) The segmented IVUS and OCT borders and the annotated vectors indicating the origin of side branches are placed perpendicularly onto the centerline at equidistant points. (3) The relative axial twist of the intravascular imaging data is then estimated using the sequential triangulation algorithm, which constitutes a simplification of the Frenet-Serret formula.¹⁵ (4) The output of these steps is a backbone of the lumen and vessel surface on IVUS and of the lumen and stent surface in OCT. These backbones and the vectors portraying the origin of side branches are back-projected onto the angiographic images, and the first frame of the model is rotated around the centerline axis; for every rotation, the back-projected vectors indicating the origin of the side branches are compared with the origin of the side branches on angiography. The angle of the first frame at which the best matching is achieved corresponds to the correct rotational orientation of the intravascular imaging frames.¹⁵ (5) 3D-B-spline high-fidelity surfaces are fitted to the backbone of the lumen and vessel wall in IVUS to generate the corresponding lumen and vessel wall models, whereas in OCT, the same process is followed for the backbone of the lumen, and for the backbone of the stent and the lumen outside the stent, to generate the lumen model at the time of the event and the lumen model postintervention, respectively.

CVR Method

This approach includes the following steps: (1) QCA segmentation of the SOI and its side branches with a diameter >1 mm in 2 end-diastolic angiographic projections ($>25^\circ$ apart) and generation of the corresponding 3D models using the QAngioXA 3D Research Edition 1.0 software. (2) Fusion of the annotated borders and vectors portraying the origin of side branches in the IVUS and OCT images with the 3D-QCA data using the following process. First, the frames that portray the side branches are placed in the corresponding location on the main vessel model, and each one is rotated so that the origin of the side branch in the IVUS/OCT frame corresponds to the origin of the side branch in the 3D-QCA reconstructions. Then, linear interpolation is applied between the proximal edge of the SOI and the first side branch, between the side branches—when >1 side branch is present—and between the last side branch and the distal edge of the SOI to estimate the location and absolute orientation of the in-between frames. (3) The 3D-QCA model of the main vessel is then removed, and (4) the lumen borders on IVUS, or the stent and the lumen borders outside the stent in OCT, are fused with the 3D-QCA models of the side branches to generate a single surface of the main vessel and side branches. This step is performed in a dedicated module named QAngioXA 3D Workbench Research Edition 1.0 (Medis Medical Imaging Systems, Leiden, the Netherlands) that allows the generation of a smooth lumen surface and seamless merging of the main vessel and the side branches at the bifurcation sites.

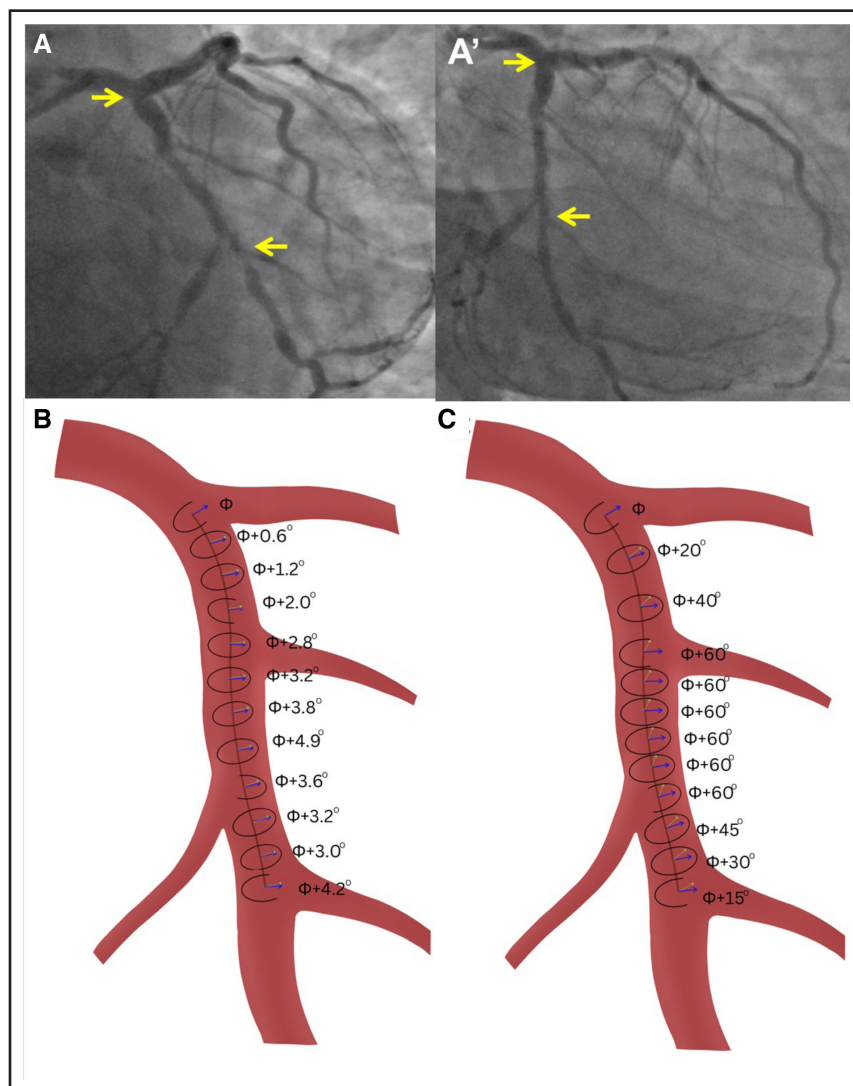


Figure 1. Case example of a vessel reconstruction using the single-vessel reconstruction (SVR) and complete vessel reconstruction (CVR) methodologies.

The **top (A and A')** shows 2 angiographic projections of a left circumflex artery, and the start and end of the reconstructed segment are indicated with yellow arrows. The **bottom left (B)** shows the 3-dimensional (3D) geometry reconstructed with the SVR methodology. In this method, each intravascular ultrasound (IVUS) contour is placed perpendicularly onto the vessel centerline at equidistant points, and the estimated relative axial twist by the sequential triangulation algorithm is shown. The **bottom right (C)** shows the 3D geometry reconstructed by the CVR methodology. Here, the IVUS contour that portrays the side branches is placed in the corresponding location and rotated to match the origin of the side branches. Linear interpolation is used to estimate the location and orientation of the contours in between the side branches, as shown. This approach resulted in larger relative axial twist and variable distances between contours.

The interobserver reproducibility of the 2 reconstruction methods was tested in 20 randomly selected vessels that were reconstructed with both approaches by 2 experienced operators; the first performed the reconstructions twice to also report the intraobserver variability.

Computational Fluid Dynamics Analysis

To ensure a smooth transition and a fully developed flow when it enters the reconstructed vessel, fixed-length extensions of 5 mm were added to the inlet and outlet of each model. The lumen and external elastic membrane models reconstructed with the 2 approaches were then meshed using Ansys (ICEM CFD 21.0 [Integrated Computer Engineering and Manufacturing, Computational Fluid Dynamics], Ansys, Canonsburg, PA), which generated a tetrahedral volume mesh with 5 layers of prism elements at the vessel wall. The lumen models reconstructed by the IVUS data acquired at baseline, and the models that corresponded to the lumen surface at baseline in the stented segments, were then imported into Ansys CFX 21.0 to perform pulsatile flow simulations and calculate the time-averaged WSS (TAWSS) and multidirectional WSS by solving the Navier-Stokes equations.

Blood was assumed to be non-Newtonian with its rheology being described by the Quemada model; this approach has been found to provide more accurate estimation of the WSS, especially in segments exposed to a low WSS environment.¹⁶ Blood density was assumed to be 1060 kg/m³, and its viscosity varied depending on shear rate and hematocrit.¹⁷ The arterial wall was considered to be rigid with no-slip conditions applied to the lumen surface, whereas zero pressure conditions were imposed at the outlet. The mean blood flow velocity was estimated as the ratio of model length×cine frame rate on angiography/average number of frames required for the contrast media to pass through the reconstructed segment. The patient-specific velocity values were combined with generic velocity waveforms reported in the literature to account for the differences between the phasic coronary blood flow in the left and right coronary arteries.^{18,19} The flow division between the main branch and side branches was calculated according to the model proposed by van der Giessen et al²⁰ The simulation was performed over 3 full cardiac cycles. The first 2 cycles were used for initialization, and the third to compute the WSS. Each cardiac cycle was divided into 100-time steps.

In each model the TAWSS, which is the WSS averaged over a cardiac cycle, the oscillatory shear index (OSI) introduced to

assess flow reversal in pulsatile flow during the cardiac cycle, the relative residence time (RRT) which represents the relative time that a blood particle resides at a distinct location at the arterial wall, the transverse WSS (TransWSS) that reflects the amount of flow perpendicular to the main direction of flow and the cross-flow index (CFI) which is the TransWSS normalized by the instantaneous WSS were computed as previously described.²¹

Data Postprocessing

The 2 reconstruction methodologies have differences in the steps that are followed for the generation of the 3D models, and in particular, the position of the borders detected on intravascular images onto the lumen centerline and in the estimation of their rotational orientation. This is likely to affect the 3D model geometry and the longitudinal and circumferential distribution of the plaque. To quantify these differences, we compared the rotational orientation of the intravascular imaging data in the models derived by the 2 methods. Moreover, we estimated the difference in the location of each border onto the lumen centerline in the models reconstructed by the 2 methods.

The IVUS-based reconstructions were split into 3-mm segments, and for each segment the following variables were estimated: the mean lumen, external elastic membrane, plaque area, and plaque burden (PB, calculated as: $100 \times \text{mean plaque area} / \text{mean external elastic membrane area}$), and NC area and its percentage, derived from IVUS–virtual histology and NIRS-IVUS. Atherosclerotic disease progression was defined in the 3-mm segments as a reduction of the lumen area and an increase in PB at follow-up.

In the OCT-based models analysis focused only in the stented segments; these were split in 1.5-mm segments, and for each segment, the mean lumen area, the stent area (that corresponds to the baseline lumen area) and the neointima area and burden (calculated as: $100 \times \text{mean neointima area} / \text{mean stent area}$) were estimated and the presence of neoatherosclerotic tissue, defined by the presence of NC or calcific tissue in the neointima, was detected.

In the 3- and 1.5-mm segments defined in the baseline IVUS and OCT models, respectively, the minimum predominant TAWSS was computed and defined as the minimum averaged TAWSS in an arc of 90° .^{3,4} Moreover, the OSI, RRT, TransWSS, and CFI at the 90° area with the minimum predominant TAWSS were calculated. The computation of the WSS metrics in 3-mm segments in the IVUS cohort and in 1.5-mm segments in the OCT cohort is based on published literature supporting the value of the hemodynamic forces measured in these segments in predicting disease progression in native and stented segments, respectively.^{2–4,11,22,23}

In the IVUS models, the first and last 3-mm segments and in the OCT models, the first and last 1.5-mm segments of the stent were excluded from the analysis to minimize the effect of the transition from model extension to the realistic geometry and the effect of the transition from the lumen to the stent contour, respectively.

Sample Size

The study was powered to demonstrate that the SVR and CVR provide different estimations for the TAWSS. In the study of Li et al⁵ the mean difference for the WSS estimations of the CVR

and SVR was 4.64 Pa. However, this report included patients with flow-limiting disease, where the WSS values are expected to be high. In patients without significant stenoses, this difference is expected to be smaller; we assumed that this will be 1.16 Pa (25% of the mean difference reported in the study of Li et al⁵) and that the SD between the TAWSS estimations of the SVR and CVR will be 2.00 Pa. Based on these assumptions, a paired *t* test indicates that we will need two hundred thirty-four 3-mm segments to demonstrate with a 2-sided α of 0.05 and 80% power that the TAWSS estimations of the SVR are different from the CVR. Considering that our data are not normally distributed and the slightly lower efficiency of the Wilcoxon signed-rank test, which retains about 95% of the power of a paired *t* test, the required number increased to two hundred fifty-two 3-mm segments. After accounting for the multilevel correlation structure (design effect of 0.3), a final target of three hundred thirty-three 3-mm segments was considered sufficient to ensure adequate statistical power. If each arterial segment has 9 3-mm segments, as noted in our previous report, then the total number of vessels required to have three hundred thirty-three 3-mm segments is estimated at 37; in this analysis, we increased this by 3 vessels to 40 vessels.⁴

Statistical Analysis

This study used a paired, within-segment analytical design, in which WSS metrics derived from SVR and CVR were computed for identical coronary segments. This design enabled direct comparison of the prognostic performance of the 2 reconstruction methodologies while controlling for anatomic and patient-level confounders. Continuous variables are presented as median (interquartile range) and categorical variables as absolute values and percentages. Comparison of the geometric characteristics of the SVR and CVR models and of the WSS estimated in the SVR and CVR reconstructions was performed using the Wilcoxon signed-rank test.

Three analytical domains were evaluated to examine the prognostic value of WSS metrics computed in the SVR and CVR models to predict changes in plaque pathology: (1) prediction of follow-up lumen area, plaque area, PB, and NC area in IVUS-based reconstructions, with all models adjusted for their corresponding baseline values to isolate progression independent of initial disease state, (2) prediction of atherosclerotic disease progression, defined using composite binary outcomes (lumen area reduction and PB increase, with sensitivity analyses incorporating NC progression), and (3) prediction of neointimal area and neointimal burden in OCT-based reconstructions of stented segments.

Statistical modeling was conducted in R using the rms package version 1.1–2.²⁴ For binary outcomes, Bayesian logistic regression models were used. For continuous and ordinal outcomes, Bayesian proportional odds models were used, providing a robust semiparametric framework that is invariant to monotonic transformations of the outcome variable and does not require distributional assumptions.²⁵ To account for the hierarchical structure of the data, all models included a random intercept for the vessel identifier, specified via the `cluster()` term in the model formula, thereby appropriately partitioning variance attributable to within-vessel correlation. Weakly informative Dirichlet priors were specified for model intercepts, with concentration parameters determined by the number of outcome categories, allowing posterior inference

to be predominantly data-driven. Regression coefficients were assigned default normal priors centered at zero. For each WSS metric, reconstruction method-specific effects were estimated using interaction terms, enabling formal quantification of differences between CVR- and SVR-derived predictors.

The superiority of the CVR over the SVR was assessed by examining the posterior distribution of the contrast $\Delta\beta = \beta_{\text{CVR}} - \beta_{\text{SVR}}$. Inference was based on 2 complementary Bayesian metrics: (1) the posterior probability of direction, $P(\Delta > 0 \text{ or } \Delta < 0)$, that was calculated to quantify the certainty that the effect differs from 0 in a consistent direction and serves as a direct measure of evidence for a difference between methods, and (2) the region of practical equivalence (ROPE) that was defined as $\Delta \in [-0.10, +0.10]$, and the probability $P(\Delta \in \text{ROPE})$ that was computed to quantify the certainty that the 2 methods yield practically equivalent predictive associations.

Superiority of CVR was established based on the following evidence framework. Strong superiority was concluded when there was high certainty of a directional difference ($P(\Delta > 0 \text{ or } \Delta < 0) \geq 0.95$) combined with a low probability of practical equivalence ($P(\Delta \in \text{ROPE}) < 0.10$). Moderate superiority was concluded when there was moderate certainty of a directional difference ($0.90 \leq P(\Delta > 0 \text{ or } \Delta < 0) < 0.95$) combined with a low probability of practical equivalence ($P(\Delta \in \text{ROPE}) < 0.10$). Practical equivalence was concluded when there was high certainty that any difference is not practically meaningful ($P(\Delta \in \text{ROPE}) \geq 0.50$). Finally, results were classified as indeterminate when there was insufficient evidence for a directional difference or superiority.

Atherosclerotic progression was predicted using multivariable models that incorporated all WSS metrics. Model performance was assessed using the explained variation on the 0 to 1 scale, which quantifies the overall predictive power of the model, analogous to an R^2 statistic. Out-of-sample predictive accuracy was evaluated using the leave-one-out information criterion, with lower values indicating superior model performance. Discrimination was assessed using Somers' Dxy rank correlation, from which the concordance statistic (C-index; $C = \text{Dxy}/2 + 0.5$) was derived. Overall model accuracy and calibration were evaluated using the Brier score, which quantifies the mean squared difference between predicted probabilities and observed outcomes.^{26,27} All credible intervals represent 95% highest posterior density intervals derived from 4000 posterior draws across 4 Markov chains using the Stan backend.

RESULTS

IVUS Models

Studied Patients

In the IBIS-4 study, data from 50 patients ($n=97$ vessels) were available for analysis, and out of these, 17 patients ($n=19$ vessels) had side branch(es) >1 mm in the SOL. In the PACMAN-AMI study, we randomly selected 15 patients ($n=21$ vessels) who were randomized to the statin group to generate a study cohort of 32 patients (40 vessels). The IBIS-4 and PACMAN-AMI study flowchart is shown in Figure 2. The mean age of the

studied population was 56 (50–62) years, and 93.8% were males. The baseline characteristics of the included patients and the studied vessels are shown in Table 1.

Comparison of the Reconstructed Models

Of the 40 IVUS vessels, follow-up data were available in 39 vessels, as 1 patient in the IBIS-4 study suffered a myocardial infarction before the 1-year follow-up and did not have a repeat IVUS assessment. The mean length of the reconstructed models was 42.8 (30.6–51.3) mm; 36 models had a single side branch, and 4 models had 2 branches. In the SVR, the IVUS contours were placed at equidistant points across the lumen centerline. Conversely, in the CVR, there was a longitudinal displacement of the IVUS contours, so the contours portraying side branches can co-localize with their origin in the 3D-QCA models; this was estimated at -0.5 ± 3.0 mm (range, -9.3 to 6.0 mm) in the baseline and -0.7 ± 2.6 mm (range, -7.4 to 3.9 mm) in the follow-up models. Significant differences were also reported in the relative rotational orientation of the IVUS contours. In the SVR method, the mean relative twist was $-0.01 \pm 0.63^\circ$ while in the CVR method, the mean torsion was $0.48 \pm 1.41^\circ$ ($P < 0.001$, Figure S1).

Reproducibility Analysis

The intraobserver and interobserver variability of the rotational orientation of the IVUS images were $-2.65 \pm 18.03^\circ$ and $4.03 \pm 18.02^\circ$ in the SVR methodology, respectively, and $-6.46 \pm 23.48^\circ$ and $-6.96 \pm 25.43^\circ$, respectively, in the CVR. The SVR methodology does not assess longitudinal displacement, while for the CVR methodology, this was calculated at 0.04 ± 0.13 mm when the same observer did the analysis twice, and at 0.00 ± 0.20 mm when the same analysis was performed by 2 different observers.

Comparison of WSS Distribution

In total, four hundred sixty-four 3 mm segments were included in the final analysis. The minimum predominant TAWSS, OSI, RRT, TransWSS, and CFI values were higher in the SVR approach compared with the corresponding estimations in the CVR models (Table 2). A case example is illustrated in Figure 3.

Association of Shear Stress With Lumen, Plaque Dimensions, and Composition at Follow-Up

The Bayesian univariable analyses results adjusted for baseline measurements are summarized in Table 3. Low TAWSS, high OSI, RRT, and TransWSS estimated by the CVR were predictors of a smaller lumen area at follow-up, whereas in the SVR, none of the WSS metrics predicted the follow-up lumen area. The TAWSS,

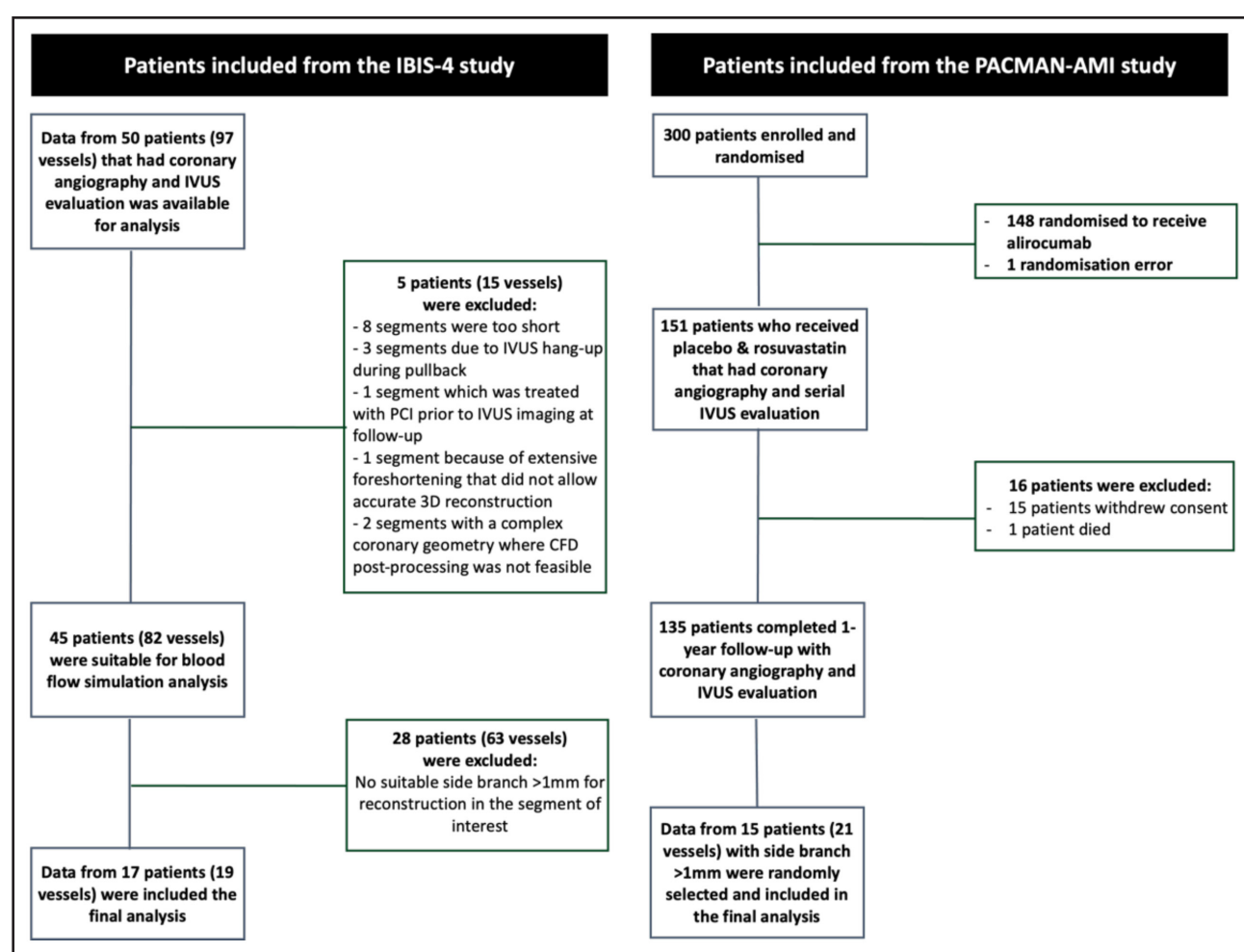


Figure 2. Flowchart of patients included from the IBIS-4 (Integrated Biomarkers and Imaging Study-4) and PACMAN-AMI (Effects of the PCSK9 Antibody Alirocumab on Coronary Atherosclerosis in Patients With Acute Myocardial Infarction) studies. IVUS indicates intravascular ultrasound; and PCI, percutaneous coronary intervention.

OSI, and RRT estimations of the CVR were strongly superior with a high probability of practical relevance to the corresponding SVR variables in predicting the lumen area at follow-up. For the TransWSS, the CVR estimations were also strongly superior, but the probability of practical relevance of this finding was low, whereas, for the CFI, the SVR demonstrated moderate superiority with a high practical relevance in predicting the follow-up lumen area compared with the CVR estimations.

Low baseline TAWSS and high OSI, RRT, and TransWSS in the CVR and low TAWSS and high TransWSS were predictors of an increase in the plaque area at follow-up. Comparison of the predictive value of these metrics demonstrated that the TAWSS of the CVR had a moderately higher predictive value with low practical relevance, whereas the OSI of the CVR demonstrated a strong superiority with a high clinical relevance in estimating the plaque area at follow-up compared with the corresponding estimations of the SVR. For the RRT, TransWSS, and CFI, the CVR method was found not to

be superior to the SVR in predicting the plaque area at follow-up.

Low TAWSS and high OSI, RRT, and TransWSS in the CVR, and none of the WSS metrics in the SVR were predictors of an increase in PB at follow-up. Comparison of the predictive value of these metrics derived by the 2 approaches showed that the TAWSS of the CVR had strong directional evidence of superiority with low practical relevance, the OSI had a strong superiority with high practical relevance, and the RRT had a moderate superiority with high practical relevance in predicting the PB at follow-up compared with the corresponding metrics of the SVR. The TransWSS and the CFI outputs of the CVR were found not to be superior to the SVR estimations in predicting the PB at follow-up.

Finally, high OSI, RRT, TransWSS, and CFI in the CVR and OSI, RRT, and TransWSS in the SVR were related to an increase in the NC component at follow-up. The estimations of the CVR were found not to be superior to the estimations of the SVR in predicting the NC at follow-up.

Table 1. Baseline Demographics of the Studied Patients and Vessels Included in the Analysis

	IVUS cohort (n=32)	OCT cohort (n=13)
Age, y	56 (50–62)	61 (57–66)
Sex (male), n (%)	30 (93.8%)	8 (61.5%)
Comorbidities, n (%)		
Diabetes	1 (3.1%)	5 (38.5%)
Hypertension	11 (34.4%)	10 (76.9%)
Hypercholesterolemia	21 (65.6%)	8 (61.5%)
Previous myocardial infarction	0 (0.0%)	6 (46.2%)
Previous PCI	0 (0.0%)	13 (100.0%)
Studied vessels, n (%)		
Left anterior descending artery	16 (40.0%)	8 (61.5%)
Left circumflex artery	16 (40.0%)	3 (23.1%)
Right coronary artery	8 (20.0%)	2 (15.4%)

IVUS indicates intravascular ultrasound; OCT, optical coherence tomography; and PCI, percutaneous coronary intervention.

Prediction of Disease Progression

Low TAWSS and high OSI and TransWSS in the CVR and low TAWSS in the SVR were associated with atherosclerotic disease progression—defined as a reduction in the lumen area and an increase in the PB at follow-up (Table 4). The TAWSS and CFI estimated in the CVR were not superior to the corresponding metrics in the SVR in predicting disease progression. The OSI derived from CVR was strongly superior to the SVR estimations with a high practical relevance, while the RRT and TransWSS of the CVR had a moderate superiority with a low practical relevance in predicting plaque progression at follow-up. In multivariable models incorporating all 5 WSS metrics, the CVR-based model outperformed the SVR-based model, with lower out-of-sample error (leave-one-out information criterion: 429 versus 551), higher

Table 2. Comparison of TAWSS and Multidirectional Shear Stress Variables in the IVUS 3-mm Segments at Baseline Models Reconstructed Using the SVR and Complete Vessel Reconstruction by Using the Wilcoxon Signed-Rank Test

	SVR, median (IQR)	CVR, median (IQR)	P value
Minimum predominant TAWSS, Pa	1.58 (0.91–2.71)	1.09 (0.62–1.96)	<0.001
OSI at the minimum TAWSS	0.00 (0.00–0.02)	0.00 (0.00–0.01)	<0.001
RRT at the minimum TAWSS, Pa ⁻¹	0.93 (0.48–1.82)	0.79 (0.44–1.47)	<0.001
TransWSS at the minimum TAWSS, Pa	0.07 (0.04–0.11)	0.04 (0.02–0.08)	<0.001
CFI at the minimum TAWSS	0.06 (0.03–0.12)	0.04 (0.02–0.08)	<0.001

CFI indicates cross-flow index; CVR, complete vessel reconstruction; IQR, interquartile range; IVUS, intravascular ultrasound; OSI, oscillatory shear index; RRT, relative residence time; SVR, single-vessel reconstruction; TAWSS, time-averaged wall shear stress; and TransWSS, transverse wall shear stress.

discrimination (C statistic: 0.725 versus 0.651), improved calibration (Brier score: 0.172 versus 0.226), and greater explained variation (27.8% versus 20.7%).

OCT Models

Thirteen patients presented with an acute coronary syndrome attributed to a neoatherosclerotic lesion that had at least 1 side branch with diameter >1 mm within the stented segment were included in the analysis; 8 models had a single side branch, and the remaining had 2 branches. The study flowchart is shown in Figure S2, whereas the baseline characteristics of the studied population and the studied vessels are presented in Table 1. The comparison of the output of the coregistration algorithms, of the WSS distribution in the SVR and CVR models, and the association of the baseline WSS metrics with the presence of neoatherosclerotic lesions are presented in the Supplemental Material.

Coronary artery reconstruction and blood flow simulation in models reconstructed using the SVR method require ≈9.42 hours per vessel, respectively, compared with ≈19.75 hours per vessel in the CVR models using a workstation with 36 cores, 128GB RAM, Intel Xeon E5-2640 @2.6GHz, Intel Corporation, Santa Clara, CA.

DISCUSSION

The present study evaluated the distribution of the TAWSS and multidirectional WSS indices in native and stented segments reconstructed using the SVR and the CVR method, and examined the efficacy of the computed WSS metrics in predicting disease progression. We found that (1) there are significant geometric differences in the reconstructed models; the CVR method requires longitudinal displacement of the IVUS and OCT contours in the lumen centerline and higher degree of rotation between contours for accurate coregistration with the origin of the side branch in 3D-QCA compared with the SVR, (2) the incorporation of side branches resulted in lower TAWSS and alters the multidirectional WSS distribution, (3) TAWSS and multidirectional WSS enabled more accurate prediction of native vessel disease progression in the CVR methodology and (4) in stented segments the WSS variables computed by the CVR was superior to those computed by the SVR in predicting neointima area and burden.

Two previous reports have examined the impact of side branches on flow patterns. Li et al,⁵ who introduced for the first time a method that incorporated the side branches in vessel geometry, demonstrated that the inclusion of side branches resulted in a reduction of the mean WSS by 4.64 Pa. However, this study included lesions with advanced disease, and, therefore, it is likely these findings do not apply to segments with mild stenoses. Similar findings were reported by Giannopoulos et al,⁶ who assessed

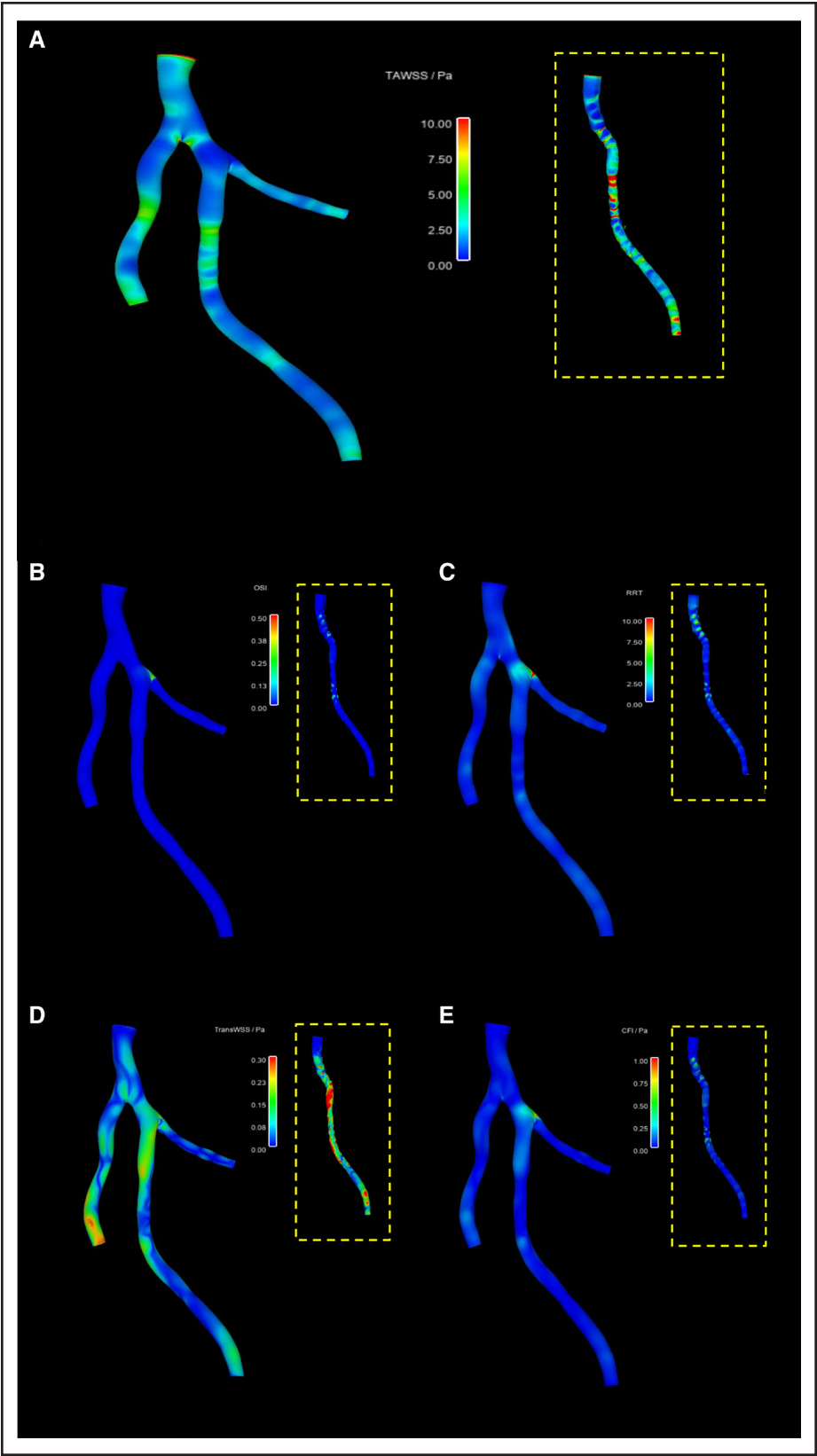


Figure 3. Comparison of wall shear stress (WSS) distribution. A case example (intravascular ultrasound [IVUS]–based reconstruction) highlighting the differences in the distribution of time-averaged WSS (TAWSS; **A**), oscillatory shear index (OSI; **B**), relative residence time (RRT; **C**), transverse WSS (TransWSS; **D**), and cross-flow index (CFI; **E**) reconstructed using the single-vessel reconstruction and complete vessel reconstruction methodologies.

Table 3. Univariate Analysis Between Lumen Area, Plaque Area, Plaque Burden, and Necrotic Core Area and WSS Metrics Computed in Models Reconstructed by the CVR and SVR Methodology

Outcome	Predictor	SVR Beta (%95% CrI)	CVR Beta (%95% CrI)	Delta (%95% CrI)	P($\Delta > 0$ or $\Delta < 0^*$)	P(Δ between -0.1 and $+0.1$)	C stat
Lumen area	Minimum predominant TAWSS, Pa	−0.01 (−0.11 to 0.08)	0.26 (0.11 to 0.41)	0.27 (0.11 to 0.43)	100	0.02	0.876
	OSI at the minimum TAWSS	−0.55 (−5.00 to 3.78)	−11.07 (−16.46 to −5.91)	−10.46 (−17.30 to −4.24)	0.99	0.00	0.875
	RRT at the minimum TAWSS, Pa ^{−1}	−0.73 (−2.57 to 1.15)	−3.59 (−5.61 to −1.41)	−2.87 (−5.51 to −0.17)	0.98	0.01	0.875
	TransWSS at the minimum TAWSS, Pa	−0.06 (−0.18 to 0.05)	−0.20 (−0.31 to −0.10)	−0.14 (−0.29 to 0.02)	0.96	0.32	0.875
	CFI at the minimum TAWSS	−1.98 (−4.71 to 0.81)	1.48 (−2.45 to 5.35)	3.44 (−0.89 to 7.64)	0.94	0.01	0.875
Plaque area	Minimum predominant TAWSS, Pa	−0.11 (−0.20 to −0.03)	−0.23 (−0.37 to −0.08)	−0.11 (−0.27 to 0.04)	0.92	0.43	0.856
	OSI at the minimum TAWSS	2.35 (−1.76 to 6.41)	10.18 (5.04 to 15.52)	7.96 (1.44 to 14.57)	0.99	0.00	0.857
	RRT at the minimum TAWSS, Pa ^{−1}	1.63 (−0.15 to 3.45)	3.27 (1.17 to 5.37)	1.68 (−1.11 to 4.45)	0.88	0.03	0.856
	TransWSS at the minimum TAWSS, Pa	0.16 (0.07 to 0.25)	0.16 (0.05 to 0.27)	−0.00 (−0.14 to 0.14)	0.52	0.84	0.856
	CFI at the minimum TAWSS	−0.62 (−3.37 to 2.07)	0.59 (−3.13 to 4.39)	1.19 (−2.94 to 5.57)	0.73	0.03	0.855
Plaque burden	Minimum predominant TAWSS, Pa	−0.03 (−0.12 to 0.06)	−0.21 (−0.35 to −0.08)	−0.18 (−0.33 to −0.04)	0.99	0.14	0.819
	OSI at the minimum TAWSS	0.34 (−4.00 to 4.56)	9.11 (4.06 to 14.47)	8.80 (2.25 to 15.21)	0.99	0.00	0.820
	RRT at the minimum TAWSS, Pa ^{−1}	0.86 (−0.94 to 2.70)	2.96 (0.88 to 5.11)	2.13 (−0.71 to 4.87)	0.93	0.02	0.819
	TransWSS at the minimum TAWSS, Pa	0.09 (−0.01 to 0.19)	0.15 (0.05 to 0.25)	0.06 (−0.07 to 0.20)	0.80	0.70	0.820
	CFI at the minimum TAWSS	1.00 (−1.90 to 3.84)	−0.42 (−4.40 to 3.67)	−1.39 (−5.94 to 3.19)	0.73	0.03	0.819
Necrotic core area	Minimum predominant TAWSS, Pa	−0.05 (−0.15 to 0.05)	−0.05 (−0.20 to 0.10)	0.00 (−0.16 to 0.16)	0.51	0.78	0.808
	OSI at the minimum TAWSS	5.59 (0.55 to 10.45)	6.52 (1.82 to 11.24)	0.93 (−5.42 to 7.61)	0.61	0.02	0.804
	RRT at the minimum TAWSS, Pa ^{−1}	3.07 (0.91 to 5.22)	2.94 (0.92 to 4.94)	−0.13 (−3.01 to 2.65)	0.54	0.05	0.799
	TransWSS at the minimum TAWSS, Pa	0.15 (0.04 to 0.27)	0.13 (0.03 to 0.24)	−0.02 (−0.17 to 0.14)	0.59	0.78	0.795
	CFI at the minimum TAWSS	2.88 (−0.24 to 5.99)	4.53 (0.48 to 8.57)	1.62 (−2.92 to 6.23)	0.75	0.03	0.817

CFI indicates cross-flow index; CrI, credible interval; C stat, C statistics; CVR, complete vessel revascularisation; OSI, oscillatory shear index; P, practical equivalence; RRT, relative residence time; RRT, relative residence time; SVR, single-vessel revascularisation; TAWSS, time-averaged wall shear stress; TransWSS, transverse wall shear stress; and WSS, wall shear stress.

WSS in 5 hearts imaged ex vivo using computed tomography angiography and showed that the exclusion of side branches significantly overestimated WSS by 2.26 ± 3.94 Pa and that this methodology reduced the ability of the reconstructed models to predict regions with pathological low WSS. However, both studies performed steady flow simulation, and thus, they did not assess the effect of side branches on multidirectional WSS. Moreover, both reports did not have follow-up imaging to examine whether the WSS metrics computed in CVR models allow better prediction of plaque progression.

The present analysis was designed to address these limitations but also to identify differences in the geometry of models reconstructed by the 2 methods using IVUS and OCT imaging data. We found that the CVR method requires longitudinal displacement and larger rotation of the intravascular imaging borders. In the IVUS-based reconstructions, the longitudinal displacement was smaller compared with the OCT-based models. This can be explained by the fact that the IVUS images, included in vessel reconstruction, were acquired at end-diastole—that is, at the same phase of the cardiac cycle

Table 4. Efficacy of WSS Metrics Computed in Models Reconstructed Using the CVR and SVR Method to Predict Atherosclerotic Disease Progression

Outcome	Predictor	SVR; β (%95% CrI)	CVR; β (%95% CrI)	Delta (%95% CrI)	$P(\Delta > 0$ or $\Delta < 0^*$)	$P(\Delta$ between −0.1 and +0.1)	C stat
Atheroscle- rotic disease progression	Minimum predominant TAWSS, Pa	−0.23 (−0.36 to −0.10)	−0.35 (−0.52 to −0.18)	−0.12 (−0.33 to 0.09)	0.89	0.41	0.584
	OSI at the minimum TAWSS	−3.18 (−10.06 to 1.38)	11.20 (2.40 to 20.01)	14.38 (5.23 to 25.57)	0.99	0.01	0.553
	RRT at the minimum TAWSS, Pa ^{−1}	−0.12 (−2.74 to 1.65)	2.50 (−0.49 to 5.21)	2.77 (−0.54 to 6.30)	0.92	0.21	0.548
	TransWSS at the minimum TAWSS, Pa	0.03 (−0.09 to 0.14)	0.19 (0.02 to 0.37)	0.16 (−0.04 to 0.38)	0.93	0.18	0.558
	CFI at the minimum TAWSS	−0.71 (−4.49 to 2.26)	−3.47 (−8.49 to 1.00)	−2.76 (−8.25 to 2.76)	0.84	0.27	0.566

CFI indicates cross-flow index; CrI, credible interval; C stat, C statistics; CVR, complete vessel revascularisation; OSI, oscillatory shear index; P, practical equivalence; RRT, relative residence time; SVR, single-vessel revascularisation; TAWSS, time-averaged wall shear stress; TransWSS, transverse wall shear stress; and WSS, wall shear stress.

as the X-ray images, used for QCA analysis—so the longitudinal displacement of the IVUS borders in these models was only due to possible foreshortening in the reconstructed segment in X-ray angiography. Conversely, in the OCT-based reconstructions, the included contours were obtained during the entire cardiac cycle, and therefore, it is essential to correct for the relative backward-forward motion of the OCT catheter in the vessel during the cardiac cycle, which in intravascular imaging studies has been reported to be up to 5.5 mm.²⁸

We also found that the rotation of the intravascular image borders is larger in the CVR compared with the estimations of the sequential triangulation algorithm used in the SVR method. However, in silico and experimental studies have shown that the relative axial twist of the intravascular images is overestimated by the sequential triangulation algorithm, which does not take into account the friction effect of the vessel wall on the catheter shaft that is expected to be larger in the bulkier IVUS probes.²⁹ The larger relative twist of the intravascular images in the CVR method should be attributed to the fact that not all side branch origins are clearly visible in X-ray images. This can explain the marginally inferior interobserver and intraobserver reproducibility of the CVR for the rotational orientation of the IVUS frames compared with the SVR methodology, and it can affect the accuracy of the CVR in the formation of coronary bifurcation geometry.

The differences in the placement of intravascular images on the lumen centerline and rotational twist, as well as the inclusion of the side branches in CVR method, resulted in significant differences in the WSS distribution in the models reconstructed by the 2 methodologies (Figure S3). We found that the SVR approach overestimated the minimum predominant TAWSS by 0.49 Pa in the IVUS-based and by 0.65 Pa in the OCT-based reconstructions compared with the CVR method. These differences are smaller than those reported in the literature. This should be attributed to the fact that previous reports included heavily diseased segments and focused on the mean TAWSS values, whereas the present

analysis included native segments with minor atheroma or segments with a large lumen poststent implantation and we report the minimum predominant WSS in 3 mm segments.

In our analysis, we performed pulsatile flow simulation and examined, the effect of side branch reconstruction in multidirectional WSS that are instigators of plaque progression.²⁰ Coronary bifurcations influence WSS distribution, causing flow disturbances at the origin of the daughter branches and an unfavorable hemodynamic milieu that promotes atherosclerotic disease progression.¹⁰ The flow patterns in these locations cannot be accurately assessed by the models generated by the SVR approach, which does not take into account flow division in these segments, resulting in differences in the multidirectional WSS estimations. In the IVUS cohort, we found high multidirectional WSS values in these locations in the CVR, but overall, the multidirectional WSS were higher in the SVR approach. This should be attributed to differences in the reconstructed surfaces in the 2 approaches. In the SVR, generation of a high-fidelity lumen surface—that is closer to the original lumen backbone—is possible, while in the CVR this would result in errors at the bifurcation site and thus the lumen geometry was constructed using the Workbench software that allows seamless construction of the bifurcation geometry but is unable to capture details of the original lumen backbone that can influence flow patterns (Figure 1). Despite these approximations as well as the assumption regarding the flow split in the main segment and side branches, we found that the WSS computed by the CVR enabled better prediction of segments that were prone to progress at follow-up than the WSS estimations of the SVR method.

Significant differences were also noted in the WSS distribution between the estimations of the SVR and CVR in the stented segments. In this cohort, the TAWSS was lower, but in contrast to what has been reported in native segments, the OSI, RRT, and CFI were higher in the CVR. This discrepancy should be attributed to the

fact that in these models, the stent borders had a circular shape (Figure S4). In the absence of irregularities, the high-fidelity lumen surface generated in the SVR models was close to the output of the Workbench software, and therefore, the multidirectional WSS were higher in the CVR because of flow loss distal to the bifurcation and the flow disturbances at the bifurcation site. In the OCT models, WSS metrics derived by the CVR approach had a superior performance to the estimations of the SVR in predicting neointima area and burden. In addition, we found that more WSS variables in the CVR were associated with neoatherosclerotic lesion formation than the SVR WSS metrics. These findings are in line with our results in native segments despite our approximations in the reconstruction of the lumen geometry at baseline and the longer follow-up period (7.4 ± 4.8 years) and support the incorporation of the side branches in models reconstructed to examine the implications of flow patterns on vessel wall response poststent implantation.

Limitations

Several limitations of the present analysis should be acknowledged. First, although this is one of the largest studies using multidirectional WSS metrics to predict plaque progression, the number of included segments remains relatively small; therefore, it was not possible to examine the value of WSS metrics in detecting disease progression at the lesion level. Similarly, the 13 stented segments in the present analysis did not allow assessment of the effects of WSS on lipid-rich neoatherosclerotic lesion formation and macrophage accumulation. Second, our IVUS study cohort included data collected by 2 different catheters that provide different estimations about the NC component. Although there is a lack of concordance between NIRS-IVUS and IVUS-virtual histology for the NC component, prospective studies of atherosclerosis have shown that the NC estimations of both modalities are a predictor of events.^{30,31} To account for this, the effect of the IVUS probe used was included in all the prediction models. Third, in the stented segments, the OCT data poststent implantation was not available. To overcome this limitation, it was assumed that the stent borders at the time of the event corresponded to the lumen borders poststent deployment.^{11,32,33} This approach has been extensively used in the past to evaluate the effect of flow patterns on vessel wall healing. Despite this assumption, we found an association between WSS metrics and neointima area and burden. Fourth, we could not evaluate the effect of vessel foreshortening on angiography on longitudinal displacement of the IVUS or OCT contours without the availability of computed tomography data that provide accurate estimations about vessel 3D geometry. Fifth, this study was limited to the comparison of 2 well-established methodologies for SVR and CVR and did not compare all the

reconstruction methods proposed in the literature. However, we are confident that our results will apply to other methods developed for SVR and CVR, as they all follow similar steps to generate vessel geometry.³⁴ Sixth, the coronary blood flow velocity was approximated by the length of the models, the Thrombolysis in Myocardial Infarction frame count, and the cine frame rate, as Doppler wire measurements were not available.^{35,36} In line with previous studies highlighting the prognostic value of WSS measured in 3D-QCA models, the computed velocity was imposed at the inflow of the models reconstructed by the 2 approaches to extract flow patterns. This established approach was chosen in this analysis as it resulted in the same boundary conditions in the inflow of the models, allowing direct comparison of the WSS pattern and its value in predicting plaque evolution. Seventh, the small number of lesions (51 lesions in the 40 vessels which had serial intravascular imaging assessment) did not allow us to draw a safe conclusion on the effect of reconstruction methodology on lesion-level plaque progression so has been omitted from the article. Finally, flow in the studied segments was estimated by the Thrombolysis in Myocardial Infarction frame count and not by a Doppler wire. This can affect the estimated WSS metrics, especially in the models incorporating side branches where flow division was approximated. Despite these approximations, the WSS estimations of the CVR method enabled more accurate prediction of disease progression and were strongly associated with neointima characteristics in stented segments.

Conclusions

CVR and incorporation of side branches in 3D vessel geometry are feasible and result in different WSS distribution compared with the SVR approach. This method enables more accurate identification of segments prone to progress in native vessels and provides WSS estimations that are strongly related to neointima proliferation in stented segments. Therefore, although it is more time-consuming, computationally challenging, and expensive, CVR should be preferred over SVR for evaluating the implications of the local hemodynamic forces in vessel wall pathobiology in native and stented segments and for training artificial intelligence solutions that will enable real-time evaluation of flow patterns and prediction of plaques that are prone to progress and cause events.³⁷

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Disclosures

None.

Supplemental Material

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